



personal website

Daniel Keith Brattan

Theoretical physicist - statistical field theory and non-equilibrium dynamics

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Research profile

Senior Research Associate (ricercatore tipo A) at the Università degli Studi di Genova, member of the STELLAR Lab and member of INFN Sezione di Genova. I am a theoretical physicist working on non-equilibrium and statistical field theory, using effective-field-theory and holographic methods to understand transport, fluctuations and collective dynamics away from equilibrium. My current research programmes focus on quasihydrodynamics, Schwinger-Keldysh effective theory, active matter and boost-agnostic mechanics, with related work in holography and condensed-matter theory.

Academic appointments

Senior Research Associate - Ricercatore tipo A

Jun 2025 - present

Dipartimento di Fisica, Università degli Studi di Genova, Italy

- Research: quasihydrodynamic theories of condensed and active matter, exotic-particle statistical mechanics, non-equilibrium steady states and fluctuating effective theory.
- Principal Investigator: PNRR project “Further beyond hydrodynamics” (FurtherQuasi; CUP D33C25000470006; MUR funding €299,975; 2025–2028).

Assegno di ricerca

Feb 2024 - May 2025

Dipartimento di Fisica, Università degli Studi di Genova, Italy

- Quasihydrodynamic theories of condensed and active matter; electrically driven phase transitions.
- Funding: PRIN 2022A8CJP3, Italian Ministry of University and Research (MUR).

Marie Skłodowska-Curie Individual Fellow

Dec 2021 - Nov 2023

Centre de Physique Théorique, École Polytechnique, France

- Quasihydrodynamics for condensed matter systems inspired by gauge-gravity duality.
- EU Horizon 2020, QuasiHydro, grant agreement 101030915.

Postdoctoral Researcher

Nov 2018 - Nov 2021

INFN Genova / Università degli Studi di Genova, Italy

- Incoherent conductivities and spontaneously broken translation invariance in magnetotransport, including applications to $\text{Bi}_2\text{Sr}_2\text{CuO}_6$ (Bi-2201).
- Funding: INFN Scientific Initiative SFT.

Postdoctoral Researcher

Oct 2016 - Sep 2018

University of Science and Technology of China, PRC

- Scale anomalies, anisotropic scaling, discrete scale invariance and renormalisation-group methods in quantum mechanics.
- Funding: National Natural Science Foundation of China, Grant No. 11235010.

Postdoctoral Researcher

Jun 2013 - Oct 2016

Technion and University of Haifa, Israel

- Gauge-gravity duality, dynamical boundary fields, anyon physics, hydrodynamics and holographic zero sound.
- Funding included the University of Haifa President’s Postdoctoral Fellowship, Fine Scholarship and Israeli Science Foundation support.

Education

PhD in Mathematics

2008 - 2013

Durham University, UK

- Thesis: “Aspects of hydrodynamics in AdS/CMT”.
- Supervisors: Mukund Rangamani and Simon Ross. Funding: STFC Studentship.

M.Sci. Natural Sciences (Mathematics and Physics), First Class Honours

2004 - 2008

Durham University, UK

- Thesis: “Hawking radiation”. Advisor: Mukund Rangamani.
- Focus: quantum fields in curved spacetime, gravity, black holes and string theory.

Publications

29 papers and preprints. Persistent profiles: [ORCID](#), [INSPIRE](#), [arXiv](#), and [Google Scholar](#). Updated 5 October 2026.

- [1] A. Amoretti, D. K. Brattan, and J. Rongen. *When duality changes the poles: $SL(2, \mathbb{Z})$ transformations of linear response EFTs*. 2026. arXiv: 2609.19075 [[hep-th](#)].
- [2] A. Amoretti, M. Anselmi, and D. K. Brattan. *Schwinger-Keldysh effective actions for non-hydrodynamic poles and branch cuts*. 2026. arXiv: 2609.16164 [[hep-th](#)].
- [3] A. Amoretti and D. K. Brattan. *Staying positive: bounds for non-Gaussian noise in Schwinger-Keldysh effective field theory*. 2026. arXiv: 2609.12050 [[cond-mat.stat-mech](#)].
- [4] A. Amoretti, D. K. Brattan, and J. Rongen. “Linear response beyond hydrodynamic poles”. In: *JHEP* 07 (2026), p. 092. doi: [10.1007/JHEP07\(2026\)092](#). arXiv: 2512.19694 [[hep-th](#)].
- [5] A. Amoretti, M. Anselmi, and D. K. Brattan. *A new web of dualities from Majorana Fermions*. 2025. arXiv: 2511.22261 [[hep-th](#)].
- [6] A. Amoretti, D. K. Brattan, and L. Martinoia. “The Hamiltonian mechanics of exotic particles”. In: *J. Stat. Mech.* 2025.12 (Dec. 2025), p. 123201. doi: [10.1088/1742-5468/ae1572](#). arXiv: 2506.13848 [[cond-mat.stat-mech](#)].
- [7] A. Amoretti, D. K. Brattan, L. Martinoia, and J. Rongen. “Dissipative electrically driven fluids”. In: *JHEP* 12 (Dec. 2024), p. 114. doi: [10.1007/JHEP12\(2024\)114](#). arXiv: 2407.18856 [[cond-mat.stat-mech](#)].
- [8] A. Amoretti, D. K. Brattan, and L. Martinoia. “Thermodynamic constraints and exact scaling exponents of flocking matter”. In: *Phys. Rev. E* 110.5 (2024), p. 054108. doi: [10.1103/PhysRevE.110.054108](#). arXiv: 2405.02283 [[cond-mat.stat-mech](#)].
- [9] D. K. Brattan, M. Matsumoto, M. Baggioli, and A. Amoretti. “Relaxed hydrodynamic theory of electrically driven nonequilibrium steady states”. In: *Phys. Rev. Research* 6.4 (2024), p. 043097. doi: [10.1103/PhysRevResearch.6.043097](#). arXiv: 2404.05568 [[cond-mat.stat-mech](#)].
- [10] A. Amoretti, D. K. Brattan, L. Martinoia, I. Matthaiaakis, and J. Rongen. “Relaxation terms for anomalous hydrodynamic transport in Weyl semimetals from kinetic theory”. In: *JHEP* 02 (2024), p. 071. doi: [10.1007/JHEP02\(2024\)071](#). arXiv: 2309.05692 [[hep-th](#)].
- [11] A. Amoretti, D. K. Brattan, L. Martinoia, and I. Matthaiaakis. “Restoring time-reversal covariance in relaxed hydrodynamics”. In: *Phys. Rev. D* 108.5 (2023), p. 056003. doi: [10.1103/PhysRevD.108.056003](#). arXiv: 2304.01248 [[hep-th](#)].
- [12] A. Amoretti, D. K. Brattan, L. Martinoia, and I. Matthaiaakis. “Leading order magnetic field dependence of conductivities in anomalous hydrodynamics”. In: *Phys. Rev. D* 108.1 (2023), p. 016003. doi: [10.1103/PhysRevD.108.016003](#). arXiv: 2212.09761 [[hep-th](#)].
- [13] A. Amoretti, D. K. Brattan, L. Martinoia, and I. Matthaiaakis. “Non-dissipative electrically driven fluids”. In: *JHEP* 05 (2023), p. 218. doi: [10.1007/JHEP05\(2023\)218](#). arXiv: 2211.05791 [[hep-th](#)].
- [14] A. Amoretti and D. K. Brattan. “On the hydrodynamics of $(2 + 1)$ -dimensional strongly coupled relativistic theories in an external magnetic field”. In: *Mod. Phys. Lett. A* 37.21 (2022), p. 2230010. doi: [10.1142/S0217732322300105](#). arXiv: 2209.11589 [[hep-th](#)].
- [15] A. Amoretti, D. K. Brattan, N. Magnoli, L. Martinoia, I. Matthaiaakis, and P. Solinas. “Destroying superconductivity in thin films with an electric field”. In: *Phys. Rev. Research* 4.3 (2022), p. 033211. doi: [10.1103/PhysRevResearch.4.033211](#). arXiv: 2202.00687 [[cond-mat.supr-con](#)].
- [16] A. Amoretti, D. Areal, D. K. Brattan, and N. Magnoli. “Hydrodynamic magneto-transport in charge density wave states”. In: *JHEP* 05 (2021), p. 027. doi: [10.1007/JHEP05\(2021\)027](#). arXiv: 2101.05343 [[hep-th](#)].

- [17] A. Amoretti, D. Arean, D. K. Brattan, and L. Martinoia. “Hydrodynamic magneto-transport in holographic charge density wave states”. In: *JHEP* 11 (2021), p. 011. doi: [10.1007/JHEP11\(2021\)011](https://doi.org/10.1007/JHEP11(2021)011). arXiv: [2107.00519](https://arxiv.org/abs/2107.00519) [hep-th].
- [18] A. Amoretti, D. K. Brattan, N. Magnoli, and M. Scanavino. “Magneto-thermal transport implies an incoherent Hall conductivity”. In: *JHEP* 08 (2020), p. 097. doi: [10.1007/JHEP08\(2020\)097](https://doi.org/10.1007/JHEP08(2020)097). arXiv: [2005.09662](https://arxiv.org/abs/2005.09662) [hep-th].
- [19] A. Amoretti, M. Meinero, D. K. Brattan, F. Caglieris, E. Giannini, M. Affronte, C. Hess, B. Buechner, N. Magnoli, and M. Putti. “A hydrodynamical description for magneto-transport in the strange metal phase of Bi-2201”. In: *Phys. Rev. Res.* 2.2 (2020), p. 023387. doi: [10.1103/PhysRevResearch.2.023387](https://doi.org/10.1103/PhysRevResearch.2.023387). arXiv: [1909.07991](https://arxiv.org/abs/1909.07991) [cond-mat.str-el].
- [20] D. K. Brattan, O. Ovdad, and E. Akkermans. “On the landscape of scale invariance in quantum mechanics”. In: *J. Phys. A* 51.43 (2018), p. 435401. doi: [10.1088/1751-8121/aadfae](https://doi.org/10.1088/1751-8121/aadfae). arXiv: [1804.10213](https://arxiv.org/abs/1804.10213) [hep-th].
- [21] D. K. Brattan. “ $\mathcal{N} = 2$ supersymmetry and anisotropic scale invariance”. In: *Phys. Rev. D* 98.3 (2018), p. 036005. doi: [10.1103/PhysRevD.98.036005](https://doi.org/10.1103/PhysRevD.98.036005). arXiv: [1801.03098](https://arxiv.org/abs/1801.03098) [hep-th].
- [22] D. K. Brattan, O. Ovdad, and E. Akkermans. “Scale anomaly of a Lifshitz scalar: A universal quantum phase transition to discrete scale invariance”. In: *Phys. Rev. D* 97 (6 Mar. 2018), p. 061701. doi: [10.1103/PhysRevD.97.061701](https://doi.org/10.1103/PhysRevD.97.061701). arXiv: [1706.00016](https://arxiv.org/abs/1706.00016) [hep-th].
- [23] M. Baggioli and D. K. Brattan. “Drag phenomena from holographic massive gravity”. In: *Class. Quant. Grav.* 34.1 (2017), p. 015008. doi: [10.1088/1361-6382/34/1/015008](https://doi.org/10.1088/1361-6382/34/1/015008). arXiv: [1504.07635](https://arxiv.org/abs/1504.07635) [hep-th].
- [24] D. K. Brattan. “A strongly coupled anyon material”. In: *JHEP* 11 (2015), p. 214. doi: [10.1007/JHEP11\(2015\)214](https://doi.org/10.1007/JHEP11(2015)214). arXiv: [1412.1489](https://arxiv.org/abs/1412.1489) [hep-th].
- [25] D. K. Brattan and G. Lifschytz. “Holographic plasma and anyonic fluids”. In: *JHEP* 1402 (2014), p. 090. doi: [10.1007/JHEP02\(2014\)090](https://doi.org/10.1007/JHEP02(2014)090). arXiv: [1310.2610](https://arxiv.org/abs/1310.2610) [hep-th].
- [26] D. K. Brattan, R. A. Davison, S. A. Gentle, and A. O’Bannon. “Collective Excitations of Holographic Quantum Liquids in a Magnetic Field”. In: *JHEP* 1211 (2012), p. 084. doi: [10.1007/JHEP11\(2012\)084](https://doi.org/10.1007/JHEP11(2012)084). arXiv: [1209.0009](https://arxiv.org/abs/1209.0009) [hep-th].
- [27] D. Brattan, J. Camps, R. Loganayagam, and M. Rangamani. “CFT dual of the AdS Dirichlet problem : Fluid/Gravity on cut-off surfaces”. In: *JHEP* 1112 (2011), p. 090. doi: [10.1007/JHEP12\(2011\)090](https://doi.org/10.1007/JHEP12(2011)090). arXiv: [1106.2577](https://arxiv.org/abs/1106.2577) [hep-th].
- [28] D. K. Brattan and S. A. Gentle. “Shear channel correlators from hot charged black holes”. In: *JHEP* 1104 (2011), p. 082. doi: [10.1007/JHEP04\(2011\)082](https://doi.org/10.1007/JHEP04(2011)082). arXiv: [1012.1280](https://arxiv.org/abs/1012.1280) [hep-th].
- [29] D. K. Brattan. “Charged, conformal non-relativistic hydrodynamics”. In: *JHEP* 1010 (2010), p. 015. doi: [10.1007/JHEP10\(2010\)015](https://doi.org/10.1007/JHEP10(2010)015). arXiv: [1003.0797](https://arxiv.org/abs/1003.0797) [hep-th].

Research funding and awards

Research funding

- **Principal Investigator**, “Further beyond hydrodynamics” (FurtherQuasi), PNRR Missione 4, Componente 2, Investimento 1.2; MUR funding €299,975, Università degli Studi di Genova (2025–2028).
- Marie Skłodowska-Curie Individual Fellowship (€196,707.84), École Polytechnique (2021-2023).
- Fine Scholarship (approx. €1,200), University of Haifa (2012-2013).
- University of Haifa President’s Postdoctoral Fellowship (approx. €17,000), University of Haifa (2012-2013).

Awards

- Peter Rowe Memorial Prize, Durham University (2008), awarded annually on the recommendation of the Board of Examiners in Mathematical Sciences to a final-year student.

Teaching, supervision and mentoring

Supervision and mentoring

- Andrea Parodi, Master’s thesis: “Quasihydrodynamic Perturbations about Electrically Driven Steady States”, Università degli Studi di Genova (2025) - co-supervisor.
- Emanuele Falchi, Master’s final-year project: “Effective quasihydrodynamics of the probe brane in the chiral symmetry broken phase”, Università degli Studi di Genova (2024) - co-supervisor.
- Jason Oyugi, Bachelor’s thesis: “Hydrodynamic response functions and analytic properties of dispersion relations”, École Polytechnique (2023) - co-supervisor.
- Niranjana Kamath, Master’s internship: “Hydrodynamics from a Holographic Perspective”, École Polytechnique (2022) - co-supervisor.
- Omrie Ovdad, doctoral mentor through the joint project “Anisotropic scaling symmetry in quantum mechanics” (2016-2018).

Teaching

- PhD course on the AdS/CFT correspondence, Università degli Studi di Genova, academic year 2024/25 (20 hours).
- Core A Mathematics tutorials, Durham University (60 hours), including preparation of lesson plans and teaching material.
- Grading: Quantum Mechanics III/IV; General Relativity III/IV; Statistical Mechanics III/IV; Algebra and Number Theory II; Data Analysis, Modeling and Statistics (30 hours each).

Training and outreach

- “Build and activate your network”, École Polytechnique (13 Jun 2022).
- “Teaching: how to develop students’ motivation”, École Polytechnique (4-6 Apr 2022).
- European Researchers’ Night (2022, 2023); academic mentoring and examination support.

Invited talks and seminars

“Relaxed hydrodynamics (a very brief overview)”	28 Sep 2023
Hydrodynamics at all Scales, Nordita, Stockholm, Sweden - invited talk.	
“Relaxed hydrodynamics”	7 Sep 2023
CMD30/FisMat 2023, Milan, Italy - invited talk.	
“Relaxed hydrodynamics (in an electric field)”	22 Mar 2023
INFN/DIFI Theoretical Physics Seminar, University of Genoa, Italy. Seminar page .	
“Broken translation invariance for fluids in external magnetic fields; a holographic perspective”	27 May 2021
Asia Pacific Center for Theoretical Physics, South Korea - invited talk.	
“Diffusion in a magnetic field”	8 Sep 2020
HoloTube seminar network. Seminar series .	
“Strongly coupled anyons”	24 Oct 2014
Joint theory meeting between Oviedo and Santiago, Spain - invited talk.	
“Fluid/gravity at finite r”	3 Nov 2011
Rudolf Peierls Centre for Theoretical Physics seminar, University of Oxford, UK - invited talk.	
“Fluid/gravity at finite r”	27 Oct 2011
Quantum Fields and Strings Seminar, University of Cambridge, UK. Seminar record .	

Conference and workshop talks

“Fluctuation-dissipation beyond hydrodynamics: nonlinear Maxwell-Cattaneo in the Schwinger-Keldysh formalism”	18 Sep 2026
FRAME2026 - Fluctuations and Randomness in Many-body Evolution, Institut Pascal, Université Paris-Saclay, France - 20-minute contribution. Programme .	
“Hamiltonian mechanics of exotic particles”	6 Jul 2026
SigmaPhi2026, Kolymvari, Chania, Crete, Greece - conference talk. Abstract booklet .	

“Holographic driven steady states”

28 Mar 2025

Holographic Perspectives on Chiral Transport and Spin Dynamics, ECT*, Trento, Italy - workshop talk. [Programme](#).

“On conductivities in anomalous hydrodynamics”

16 Mar 2023

Holographic Perspectives on Chiral Transport and Spin Dynamics, ECT*, Trento, Italy - 20-minute workshop contribution.

“A hydrodynamic description for transport in the strange metal phase of cuprates”

9 Mar 2023

APS March Meeting, Las Vegas, USA - contributed presentation.

“Relaxed hydrodynamics with an external electric field - or the Drude model”

28 Jun 2022

Recent Developments in Strongly Correlated Quantum Matter, Stockholm, Sweden - spotlight talk.

“A holographic plasma and many anyonic fluids”

17 Dec 2013

Israel Physical Society meeting, Weizmann Institute, Israel.

“A holographic plasma and many anyonic fluids”

1 Dec 2013

Joint Seminars in Theoretical High Energy Physics, Neve Shalom, Israel.

“Charged, conformal, non-relativistic fluids from gravity”

20–24 Sep 2010

Black Holes and Higher Dimensions, Imperial College London, UK.

“Charged, conformal non-relativistic hydrodynamics”

29 Jun 2010

North British Mathematical Physics Seminar, UK.

Posters

“A hydrodynamic description for transport in the strange metal phase of cuprates”

25 Jul 2022

International Conference on Strongly Correlated Electron Systems, Amsterdam, The Netherlands.

“Scale anomaly of a Lifshitz scalar: a universal quantum phase transition to discrete scale invariance”

26 Jun 2017

Strings 2017, Tel Aviv, Israel.

Academic service

Committees and organisation

- IOP High Energy Particle Physics Committee student representative (2011-2012), representing theoretical-physics students and contributing to conference-funding decisions.
- Young Theorists’ Forum 2010 and 2011, head of organising committee; secured support including from the Institute of Physics and the Scottish Universities Physics Alliance.
- Founded the student seminar at the University of Science and Technology of China; organised the “Quantum Field Theory for Dummies” student seminar.

Refereeing: Journal of High Energy Physics; Physics Letters B; Modern Physics Letters A.